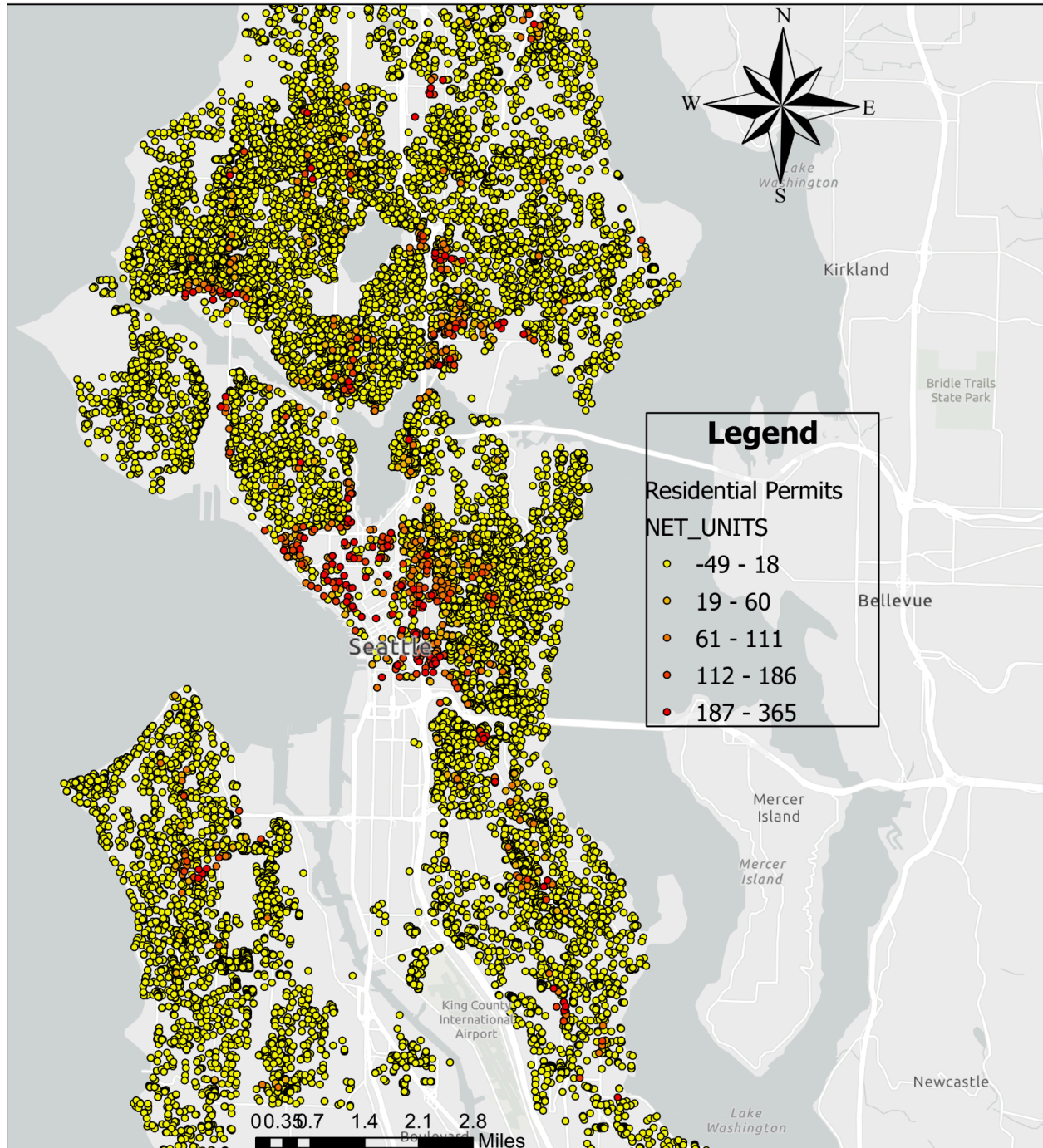
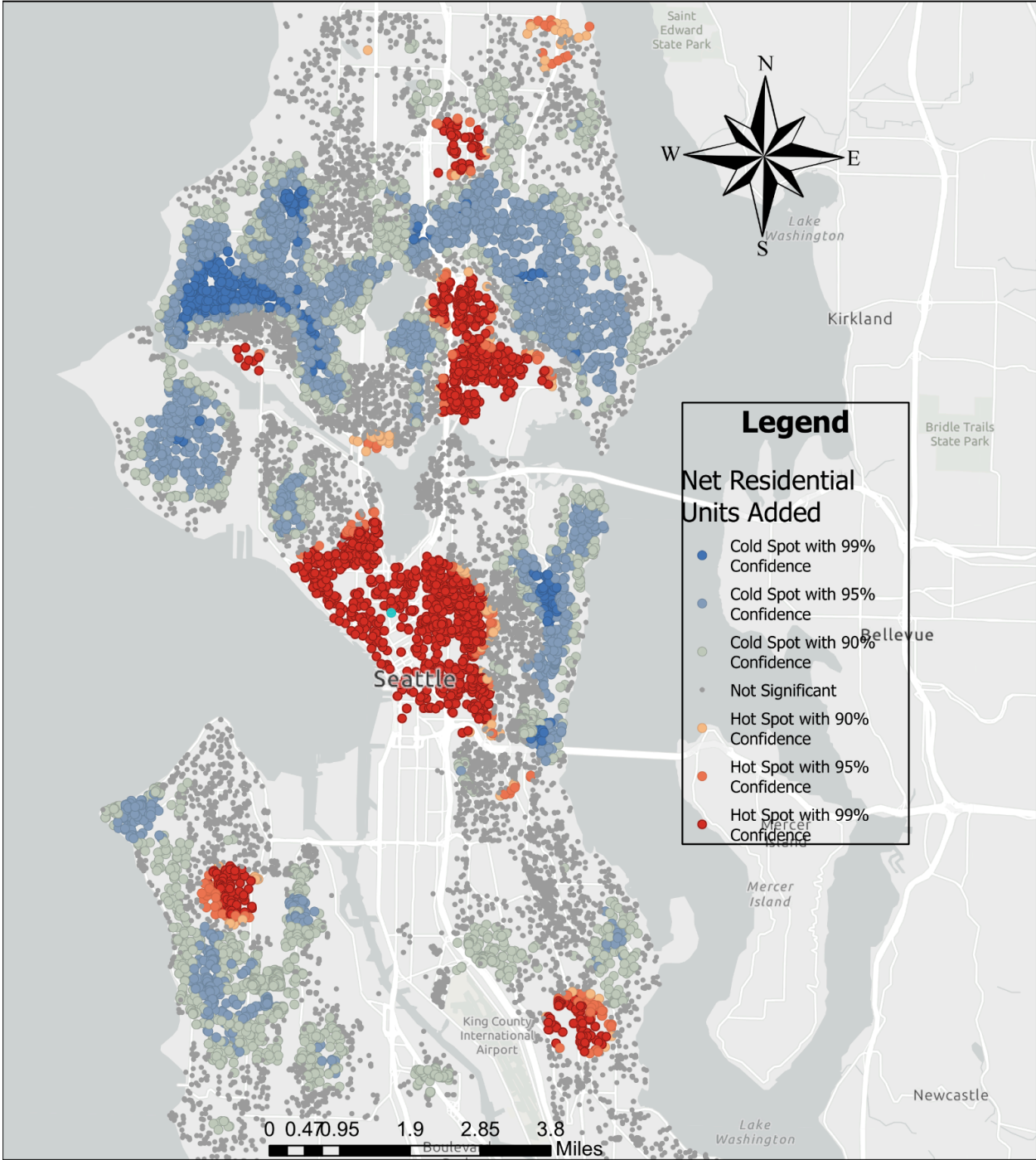


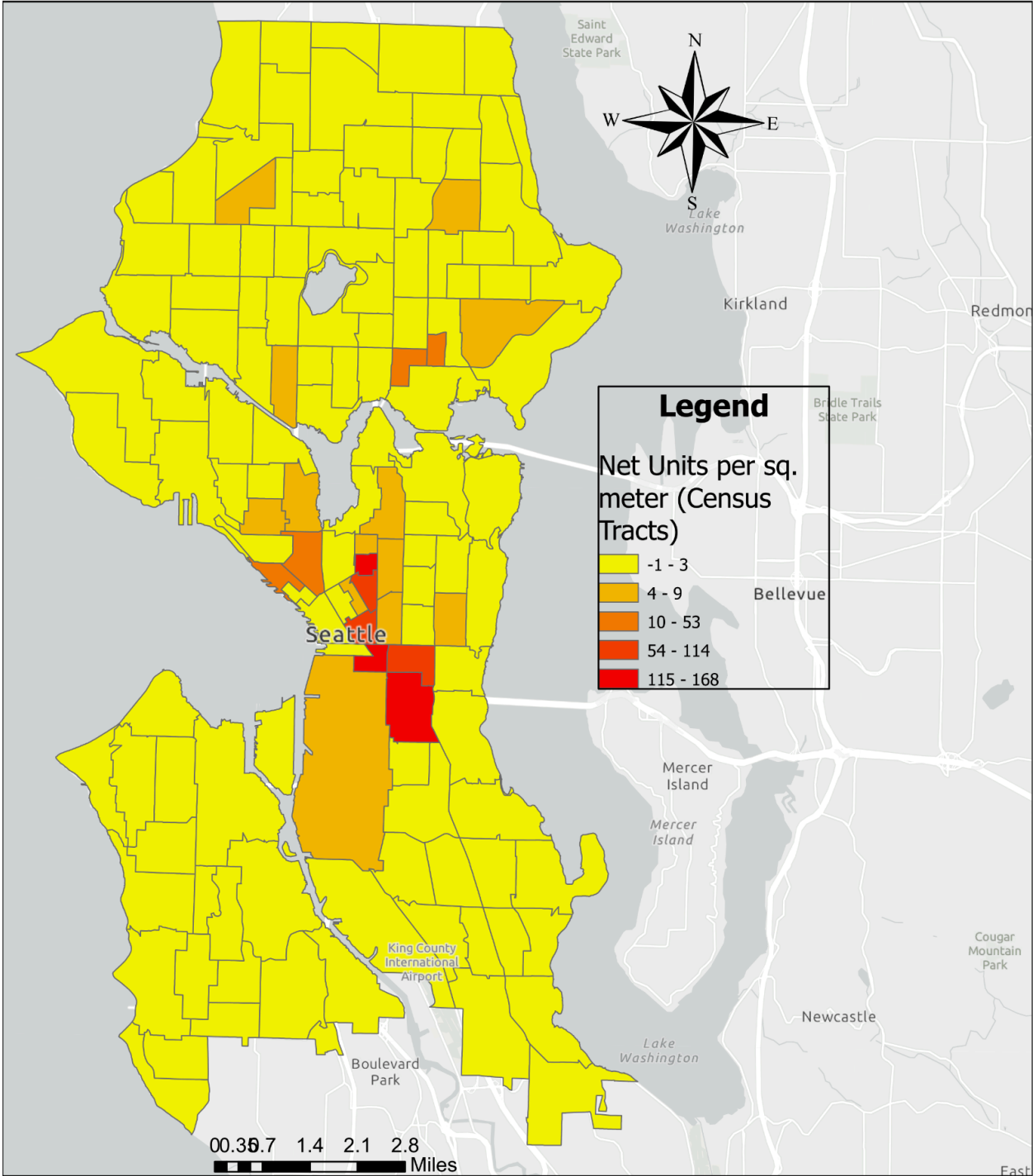
Seattle Residential Development: Construction Value per Net Unit



Hot Spot Analysis of Net Residential Units Added, Seattle 2002- 2025



Net Housing Units Built by Census Tract, Seattle 2002-2025



Report:

Seattle's residential building landscape over the past two decades reflects a city responding to powerful economic forces, demographic pressure, and shifting land use priorities. After performing multiple forms of spatial analysis on this data, the patterns that emerged revealed clear geographic concentrations and recognizable surges, while still containing some surprising outliers.

The first pattern I recognized when looking at the housing permit data of Seattle is how it reflected economic changes in its region at the time. It reflects the financial crisis in 2008, then jumps up with the Seattle tech boom of the mid 2010s. This change is also reflected in the density growth of South Lake Union and Denny Triangle in Downtown Seattle. Amazon's campus expansion in South Lake Union, placed with the arrival of dozens of supporting firms, generated housing demands that the permit records visualize easily. The influx of large unit-filled single permits built in this time is reflected in the all versions of heat maps created in this assignment.

The maps above are related in pattern but differentiate enough to provide different information. The hotspot map shows the density of each cluster through dominant red clusters. These run through Capitol Hill, the Denny Triangle, and South Lake Union before extending south through SODO. This aligns closely with light rail infrastructure and prior commercial zoning, making them target spots to fill with residential buildings. Single family neighborhoods make up the majority of the cold spots. On the other hand, the choropleth displays where the density is thickest. There are strong similarities in some census tracts, but others that seem to be hotspots on the hotspot map, read as moderate in the choropleth, serving as a reminder that raw clustering does not always translate to density in individual census tracts.

Taken together, these maps illustrate that Seattle's residential development boom was real, substantial, and geographically concentrated. The urban core absorbed the majority of growth, driven by economic demand and policy that favored density in transit-accessible areas. Understanding where that growth did and did not occur is essential context for understanding and evaluating Seattle's housing market, development, and challenges against housing affordability. The growth of Seattle as a modern city is found in these datasets, setting a foundation for the urban population seen today.